Problems

1.
   a. How many different 7-place license plates are possible if the first 2 places are for letters and the other 5 for numbers?
   b. Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.

2. How many outcome sequences are possible when a die is rolled four times, where we say, for instance, that the outcome is 3, 4, 3, 1 if the first roll landed on 3, the second on 4, the third on 3, and the fourth on 1?
3. Twenty workers are to be assigned to 20 different jobs, one to each job. How many different assignments are possible?
4. John, Jim, Jay, and Jack have formed a band consisting of 4 instruments. If each of the boys can play all 4 instruments, how many different arrangements are possible? What if John and Jim can play all 4 instruments, but Jay and Jack can each play only piano and drums?
5. For years, telephone area codes in the United States and Canada consisted of a sequence of three digits. The first digit was an integer between 2 and 9, the second digit was either 0 or 1, and the third digit was any integer from 1 to 9. How many area codes were possible? How many area codes starting with a 4 were possible?
6. A well-known nursery rhyme starts as follows:
   "As I was going to St. Ives
   I met a man with 7 wives.
   Each wife had 7 sacks.
   Each sack had 7 cats.
   Each cat had 7 kittens. . ."
   How many kittens did the traveler meet?

7.
   a. In how many ways can 3 boys and 3 girls sit in a row?
   b. In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?
   c. In how many ways if only the boys must sit together?
   d. In how many ways if no two people of the same sex are allowed to sit together?

8. When all letters are used, how many different letter arrangements can be made from the letters
   a. Fluke?
   b. Propose?
   c. Mississippi?
   d. Arrange?

9. A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible?
10. In how many ways can 8 people be seated in a row if
   a. there are no restrictions on the seating arrangement?
   b. persons A and B must sit next to each other?
   c. there are 4 men and 4 women and no 2 men or 2 women can sit next to each other?
   d. there are 5 men and they must sit next to one another?
   e. there are 4 married couples and each couple must sit together?

11. In how many ways can 3 novels, 2 mathematics books, and 1 chemistry book be arranged on a bookshelf if

a. the books can be arranged in any order?
b. the mathematics books must be together and the novels must be together?
c. the novels must be together, but the other books can be arranged in any order?

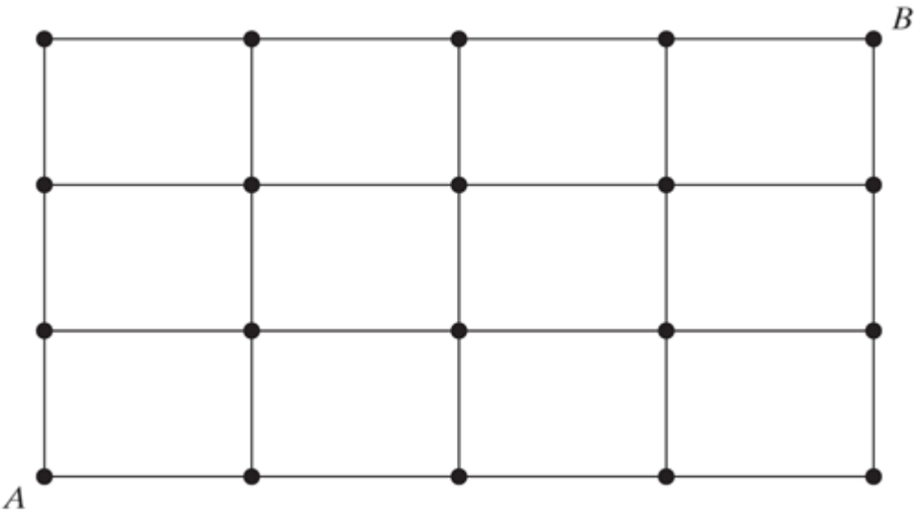
12. How many 3 digit numbers xyz, with x, y, z all ranging from 0 to 9 have at least 2 of their digits equal. How many have exactly 2 equal digits.
13. How many different letter permutations, of any length, can be made using the letters M O T T O. (For instance, there are 3 possible permutations of length 1.)
14. Five separate awards (best scholarship, best leadership qualities, and so on) are to be presented to selected students from a class of 30. How many different outcomes are possible if
a. a student can receive any number of awards?
b. each student can receive at most 1 award?

15. Consider a group of 20 people. If everyone shakes hands with everyone else, how many handshakes take place?
16. How many 5-card poker hands are there?
17. A dance class consists of 22 students, of which 10 are women and 12 are men. If 5 men and 5 women are to be chosen and then paired off, how many results are possible?
18. A student has to sell 2 books from a collection of 6 math, 7 science, and 4 economics books. How many choices are possible if
a. both books are to be on the same subject?
b. the books are to be on different subjects?

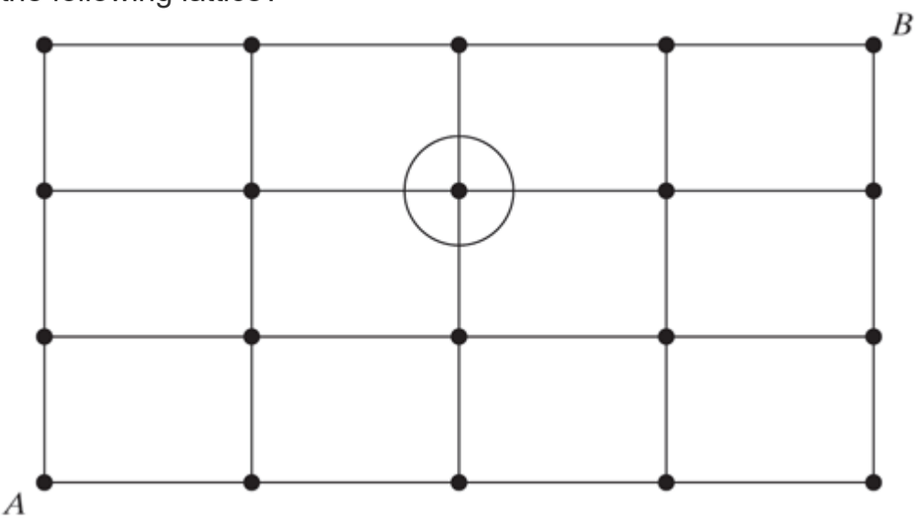
19. Seven different gifts are to be distributed among 10 children. How many distinct results are possible if no child is to receive more than one gift?
20. A committee of 7, consisting of 2 Republicans, 2 Democrats, and 3 Independents, is to be chosen from a group of 5 Republicans, 6 Democrats, and 4 Independents. How many committees are possible?
21. From a group of 8 women and 6 men, a committee consisting of 3 men and 3 women is to be formed. How many different committees are possible if
a. 2 of the men refuse to serve together?
b. 2 of the women refuse to serve together?
c. 1 man and 1 woman refuse to serve together?

22. A person has 8 friends, of whom 5 will be invited to a party.
a. How many choices are there if 2 of the friends are feuding and will not attend together?
b. How many choices if 2 of the friends will only attend together?

23. Consider the grid of points shown at the top of the next column. Suppose that, starting at the point labeled A, you can go one step up or one step to the right at each move. This procedure is continued until the point labeled B is reached. How many different paths from A to B are possible?
Hint: Note that to reach B from A, you must take 4 steps to the right and 3 steps upward.



24. In **Problem 23**, how many different paths are there from A to B that go through the point circled in the following lattice?



25. A psychology laboratory conducting dream research contains 3 rooms, with 2 beds in each room. If 3 sets of identical twins are to be assigned to these 6 beds so that each set of twins sleeps in different beds in the same room, how many assignments are possible?

26.

a. Show $\sum_{k=0}^{n} \binom{n}{k} 2^k = 3^n$
b. Simplify $\sum_{k=0}^{n} \binom{n}{k} x^k$

27. Expand $(3x^2+y)^5$.

28. The game of bridge is played by 4 players, each of whom is dealt 13 cards. How many bridge deals are possible?

29. Expand $(x_1+2x_2+3x_3)^4$.

30. If 12 people are to be divided into 3 committees of respective sizes 3, 4, and 5, how many divisions are possible?

31. If 8 new teachers are to be divided among 4 schools, how many divisions are possible? What if each school must receive 2 teachers?

32. Ten weight lifters are competing in a team weight-lifting contest. Of the lifters, 3 are from the United States, 4 are from Russia, 2 are from China, and 1 is from Canada. If the scoring takes account of the countries that the lifters represent, but not their individual identities, how many different outcomes are possible from the point of view of scores? How many different outcomes correspond to results in which the United States has 1 competitor in the top three and 2 in the bottom three?

33. Delegates from 10 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?

* 34. If 8 identical blackboards are to be divided among 4 schools, how many divisions are possible? How many if each school must receive at least 1 blackboard?

* 35. An elevator starts at the basement with 8 people (not including the elevator operator) and discharges them all by the time it reaches the top floor, number 6. In how many ways could the operator have perceived the people leaving the elevator if all people look alike to him? What if the 8 people consisted of 5 men and 3 women and the operator could tell a man from a woman?

* 36. We have $20,000 that must be invested among 4 possible opportunities. Each investment must be integral in units of $1000, and there are minimal investments that need to be made if one is to invest in these opportunities. The minimal investments are $2000, $2000, $3000, and $4000. How many different investment strategies are available if

a. an investment must be made in each opportunity?
b. investments must be made in at least 3 of the 4 opportunities?

* 37. Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.

a. How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?
b. How many outcomes are possible when 3 of the 10 fish caught are trout?
c. How many when at least 2 of the 10 are trout?

Theoretical Exercises

1. Prove the generalized version of the basic counting principle.
2. Two experiments are to be performed. The first can result in any one of m possible outcomes. If the first experiment results in outcome i, then the second experiment can result in any of ni possible outcomes, i=1, 2,. . .,m. What is the number of possible outcomes of the two experiments?
3. In how many ways can r objects be selected from a set of n objects if the order of selection is considered relevant?
4. There are (nr) different linear arrangements of n balls of which r are black and n−r are white. Give a combinatorial explanation of this fact.
5. Determine the number of vectors (x1,. . .,xn), such that each xi is either 0 or 1 and $\sum_{i=1}^{n} x_i \geq k$

6. How many vectors x1,. . .,xk are there for which each xi is a positive integer such that $1 \leq x_i \leq n$ and $x_1 < x_2 < \cdots < x_k$?
7. Give an analytic proof of **Equation (4.1)**.
8. Prove that
$\binom{n+m}{r} = \binom{n}{0}\binom{m}{r} + \binom{n}{1}\binom{m}{r-1} + \cdots + \binom{n}{r}\binom{m}{0}$

Hint: Consider a group of n men and m women. How many groups of size r are possible?
9. Use **Theoretical Exercise 8** to prove that
$\binom{2n}{n} = \sum_{k=0}^{n} \binom{n}{k}^2$

10. From a group of n people, suppose that we want to choose a committee of k, $k \leq n$, one of whom is to be designated as chairperson.
    a. By focusing first on the choice of the committee and then on the choice of the chair, argue that there are $\binom{n}{k}$ k possible choices.
    b. By focusing first on the choice of the nonchair committee members and then on the choice of the chair, argue that there are $\binom{n}{k-1}$ $(n-k+1)$ possible choices.
    c. By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are $n\binom{n-1}{k-1}$ possible choices.
    d. Conclude from parts (a), (b), and (c) that
    $k\binom{n}{k} = (n-k+1)\binom{n}{k-1} = n\binom{n-1}{k-1}$

    e. Use the factorial definition of $\binom{n}{r}$ to verify the identity in part (d).

11. The following identity is known as Fermat's combinatorial identity:
$\binom{n}{k} = \sum_{i=k}^{n} \binom{i-1}{k-1} \quad n \geq k$

Give a combinatorial argument (no computations are needed) to establish this identity.
Hint: Consider the set of numbers 1 through n. How many subsets of size k have i as their highest numbered member?
12. Consider the following combinatorial identity:
$\sum_{k=1}^{n} k\binom{n}{k} = n \cdot 2^{n-1}$

    a. Present a combinatorial argument for this identity by considering a set of n people and determining, in two ways, the number of possible selections of a committee of any size and a chairperson for the committee.
    Hint:
        i. How many possible selections are there of a committee of size k and its chairperson?
        ii. How many possible selections are there of a chairperson and the other committee members?

b. Verify the following identity for n=1, 2, 3, 4, 5:
$\sum_{k=1}^{n}(nk)\, k2=2n-2n(n+1)$

For a combinatorial proof of the preceding, consider a set of n people and argue that both sides of the identity represent the number of different selections of a committee, its chairperson, and its secretary (possibly the same as the chairperson).
Hint:
i. How many different selections result in the committee containing exactly k people?
ii. How many different selections are there in which the chairperson and the secretary are the same? (ANSWER: n2n−1.)
iii. How many different selections result in the chairperson and the secretary being different?

c. Now argue that
$\sum_{k=1}^{n}(nk)\, k3=2n-3n2(n+3)$

13. Show that, for n> 0,
$\sum_{i=0}^{n}(-1)i(ni)=0$

Hint: Use the binomial theorem.
14. From a set of n people, a committee of size j is to be chosen, and from this committee, a subcommittee of size i,i≤j, is also to be chosen.
a. Derive a combinatorial identity by computing, in two ways, the number of possible choices of the committee and subcommittee–first by supposing that the committee is chosen first and then the subcommittee is chosen, and second by supposing that the subcommittee is chosen first and then the remaining members of the committee are chosen.
b. Use part (a) to prove the following combinatorial identity:
$\sum_{j=i}^{n}(nj)(ji)=(ni)\, 2n-i \quad i\leq n$

c. Use part (a) and **Theoretical Exercise 13** to show that
$\sum_{j=i}^{n}(nj)(ji)\,(-1)n-j=0 \quad i<n$

15. Let Hk(n) be the number of vectors x1,. . .,xk for which each xi is a positive integer satisfying 1≤xi≤n and x1≤x2≤⋯≤xk.
a. Without any computations, argue that
$H1(n)=nHk(n)=\sum_{j=1}^{n}Hk-1(j) \quad k>1$

Hint: How many vectors are there in which xk=j?
b. Use the preceding recursion to compute H3(5).
Hint: First compute H2(n) for n=1, 2, 3, 4, 5.

16. Consider a tournament of n contestants in which the outcome is an ordering of these contestants, with ties allowed. That is, the outcome partitions the players into groups, with the first group consisting of the players who tied for first place, the next group being those who tied for the next-best position, and so on. Let N(n) denote the number of different possible outcomes. For instance, N(2)=3, since, in a tournament with 2 contestants, player 1 could be uniquely first, player 2 could be uniquely first, or they could tie for first
a. List all the possible outcomes when n=3.
b. With N(0) defined to equal 1, argue, without any computations, that
$N(n)=\sum_{i=1}^{n}(ni)\, N(n-i)$

Hint: How many outcomes are there in which i players tie for last place?

c. Show that the formula of part (b) is equivalent to the following:
$N(n)=\sum_{i=0}^{n-1}\binom{n}{i}N(i)$

d. Use the recursion to find N(3) and N(4).

17. Present a combinatorial explanation of why $\binom{n}{r}=\binom{n}{r, n-r}$
18. Argue that
$\binom{n}{n_1,n_2,\ldots,n_r}=\binom{n-1}{n_1-1,n_2,\ldots,n_r}+\binom{n-1}{n_1,n_2-1,\ldots,n_r}+\cdots+\binom{n-1}{n_1,n_2,\ldots,n_r-1}$

Hint: Use an argument similar to the one used to establish **Equation (4.1)**.
19. Prove the multinomial theorem.
* 20. In how many ways can n identical balls be distributed into r urns so that the ith urn contains at least m_i balls, for each $i=1,\ldots,r$? Assume that $n\geq\sum_{i=1}^{r}m_i$.
* 21. Argue that there are exactly $\binom{k}{n-r+k}$ solutions of
$x_1+x_2+\cdots+x_r=n$

for which exactly k of the x_i are equal to 0.
* 22. Consider a function $f(x_1,\ldots,x_n)$ of n variables. How many different partial derivatives of order r does f possess?
* 23. Determine the nu mber of vectors $(x_1,\ldots,x_n)$ such that each x_i is a nonnegative integer and $\sum_{i=1}^{n}x_i\leq k$

Self-Test Problems and Exercises

1. How many different linear arrangements are there of the letters A, B, C, D, E, F for which
   a. A and B are next to each other?
   b. A is before B?
   c. A is before B and B is before C?
   d. A is before B and C is before D?
   e. A and B are next to each other and C and D are also next to each other?
   f. E is not last in line?

2. If 4 Americans, 3 French people, and 3 British people are to be seated in a row, how many seating arrangements are possible when people of the same nationality must sit next to each other?
3. A president, treasurer, and secretary, all different, are to be chosen from a club consisting of 10 people. How many different choices of officers are possible if
   a. there are no restrictions?
   b. A and B will not serve together?
   c. C and D will serve together or not at all?
   d. E must be an officer?
   e. F will serve only if he is president?

4. A student is to answer 7 out of 10 questions in an examination. How many choices has she? How many if she must answer at least 3 of the first 5 questions?
5. In how many ways can a man divide 7 gifts among his 3 children if the eldest is to receive 3 gifts and the others 2 each?
6. How many different 7-place license plates are possible when 3 of the entries are letters and 4 are digits? Assume that repetition of letters and numbers is allowed and that there is no restriction on where the letters or numbers can be placed.
7. Give a combinatorial explanation of the identity
$(nr)=(nn-r)$

8. Consider n-digit numbers where each digit is one of the 10 integers 0,1,. . .,9. How many such numbers are there for which
   a. no two consecutive digits are equal?
   b. 0 appears as a digit a total of i times, i=0,. . .,n?

9. Consider three classes, each consisting of n students. From this group of 3n students, a group of 3 students is to be chosen.
   a. How many choices are possible?
   b. How many choices are there in which all 3 students are in the same class?
   c. How many choices are there in which 2 of the 3 students are in the same class and the other student is in a different class?
   d. How many choices are there in which all 3 students are in different classes?
   e. Using the results of parts (a) through (d), write a combinatorial identity.

10. How many 5-digit numbers can be formed from the integers 1,2,. . .,9 if no digit can appear more than twice? (For instance, 41434 is not allowed.)
11. From 10 married couples, we want to select a group of 6 people that is not allowed to contain a married couple.
   a. How many choices are there?
   b. How many choices are there if the group must also consist of 3 men and 3 women?

12. A committee of 6 people is to be chosen from a group consisting of 7 men and 8 women. If the